

AHAR, Iran, Islamic Republic of

WMO: 407040

Lat: **38.433N**Lon: **47.067E**Elev: **4560**StdP: **12.43**Time Zone: **3.50 (IRN)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.1	15.1	0.4	6.6	16.0	4.5	8.2	20.2	27.3	39.0	25.1	39.7	4.4	290	0.517

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	22.5	90.5	62.0	87.6	61.7	84.9	60.9	65.6	83.0	64.4	81.2	63.2	79.6	11.1	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.7	90.8	72.5	58.3	86.3	71.8	57.1	82.4	70.9	33.1	82.7	32.0	81.2	31.1	79.3	73.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.4	21.1	18.4	DB	2.6	96.9	6.6	2.8	-2.2	98.9	-6.0	100.6	-9.7	102.1	-14.5	104.2	(4)
(5)			WB	1.6	67.5	6.0	1.2	-2.7	68.4	-6.3	69.1	-9.6	69.8	-14.0	70.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.9	31.9	34.8	42.7	51.0	60.0	68.3	72.3	72.4	65.1	55.0	43.9	35.8		
	DBStd	15.92	8.60	8.20	8.91	7.72	5.17	4.49	4.68	4.53	5.79	7.35	7.73	8.69		
	HDD50	2009	561	426	256	77	2	0	0	0	1	37	205	444		
	HDD65	5109	1026	845	692	424	170	19	6	6	67	316	632	906		
	CDD50	3054	0	0	29	106	310	550	693	695	453	192	23	3		
	CDD65	681	0	0	0	3	14	119	233	236	70	6	0	0		
	CDH74	6706	0	0	1	52	197	1200	2080	2384	716	76	0	0		
	CDH80	2168	0	0	0	10	23	342	716	887	186	4	0	0		
(14) Wind	WSAvg	7.8	9.8	9.2	9.2	8.0	7.0	7.2	7.4	7.1	6.3	6.2	7.3	8.9		
(15) Precipitation	PrecAvg	12.2	0.9	0.9	1.4	1.9	2.0	0.9	0.4	0.3	0.4	1.0	1.2	0.9		
	PrecMax	16.3	1.5	2.2	2.6	3.4	3.9	2.4	1.5	1.8	1.5	3.4	3.7	2.4		
	PrecMin	8.8	0.1	0.2	0.7	0.6	0.6	0.0	0.0	0.0	0.0	0.2	0.1	0.2		
	PrecStd	2.0	0.4	0.4	0.4	0.8	0.9	0.6	0.4	0.4	0.4	0.7	0.7	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.9	56.1	71.0	80.7	82.9	92.0	95.7	94.2	89.4	80.1	66.8	60.5		
		MCWB	40.4	42.5	48.8	54.6	57.0	60.1	62.7	61.5	59.1	54.7	48.1	43.9		
	2%	DB	48.3	52.7	65.1	72.8	79.1	87.0	90.1	90.6	85.1	76.1	62.9	54.3		
		MCWB	38.1	40.3	46.5	51.7	56.1	61.0	63.6	61.7	58.1	53.7	45.9	41.5		
	5%	DB	45.7	49.4	60.8	68.5	75.7	83.9	86.8	87.8	81.7	72.2	59.4	50.4		
		MCWB	36.8	38.7	44.6	50.4	55.4	60.1	63.0	61.7	58.5	52.8	45.3	39.7		
	10%	DB	42.8	46.2	56.8	64.5	72.5	80.8	83.6	85.0	78.2	68.1	55.9	47.4		
		MCWB	34.9	37.2	43.0	48.9	54.7	59.3	62.0	61.1	58.5	51.3	44.3	38.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.1	44.0	50.0	56.2	61.1	65.2	67.9	67.1	65.0	58.1	50.8	45.3		
		MCDB	49.4	54.4	65.8	73.6	75.9	84.0	86.7	84.4	78.6	71.8	61.0	56.5		
	2%	WB	39.3	41.4	47.7	53.7	58.9	63.4	66.0	65.5	62.9	55.8	48.6	42.8		
		MCDB	46.4	50.0	62.8	68.4	73.1	80.7	84.6	81.9	76.9	69.2	57.9	51.8		
	5%	WB	37.1	39.6	45.5	51.8	57.5	61.9	64.5	64.4	61.4	54.4	46.8	40.8		
		MCDB	44.4	47.7	58.8	64.7	70.7	78.5	82.2	80.8	75.4	67.6	56.2	48.6		
	10%	WB	35.4	37.8	43.7	50.1	56.0	60.7	63.1	63.3	60.1	52.9	45.0	38.9		
		MCDB	42.1	45.5	55.5	61.9	68.6	76.7	79.8	79.5	73.4	65.0	54.2	46.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.4	14.8	17.2	19.0	20.3	22.1	20.3	22.5	21.2	19.0	16.3	14.2		
		MCDBR	14.5	18.5	23.4	26.2	25.9	27.5	25.4	28.0	28.8	26.3	22.4	17.7		
		MCWBR	8.4	10.1	10.7	10.7	8.9	9.3	9.1	10.0	10.5	10.2	11.1	10.0		
	5% WB	MCDBR	13.4	16.7	20.8	22.4	22.7	23.6	21.8	23.3	22.9	22.5	17.4	15.4		
		MCWBR	8.3	9.6	10.1	10.0	8.9	8.5	8.0	8.6	9.1	9.6	9.2	9.3		
(40) Clear-Sky Solar Irradiance	taub		0.309	0.346	0.403	0.477	0.470	0.454	0.495	0.460	0.409	0.394	0.332	0.305		
	taud		2.331	2.214	2.069	1.905	1.974	2.055	1.963	2.053	2.170	2.155	2.312	2.357		
	Ebn at Noon		273	275	269	255	260	263	251	257	264	254	261	266		
	Edh at Noon		29	37	48	60	57	53	57	51	43	39	30	26		
(44) All-Sky Solar Radiation	RadAvg		745	1020	1251	1552	1960	2382	2325	2120	1708	1179	790	647		
	RadStd		69	131	126	161	103	132	116	109	103	104	98	80		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (10 neighbors)		N/A	N/A	N/A	+1.94	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page